

Two extensions of the lower reproducing-kernel thesis for weighted composition operators

A substantial partial answer to arXiv:1812.08121

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Status. Substantial partial result, likely valid. Both source questions have a full positive answer when the composition symbol is a finite Blaschke product. For an arbitrary symbol, the finite-zero hypothesis can also be removed when the zero multiplicities admit an analytic $p/2$ -power; in particular this covers every even integer p . The unrestricted questions remain open here.

1 Source questions

The source is Chalendar and Partington, *The reproducing kernel thesis for lower bounds of weighted composition operators* [1]. Remark 3.2 on source PDF page 8 asks whether the kernel thesis extends from A^2 to the standard weighted Bergman spaces A_α^2 , $\alpha > -1$:

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we still have the equivalence of (i) and (ii) in Theorem 3.1 since the proof of Lemma 3.1 in [16] is easily seen to extend to this situation. However, at present we do not know whether the equivalence with (iii) still holds.

After Theorem 3.4, on source PDF page 9, the authors ask whether their A^p theorem extends to weights with infinitely many zeros:

It would be interesting to know whether Theorem 3.4 extends to the case when h has infinitely-many zeros, and the corresponding contractive divisor G may not be bounded.

Write $dA_\alpha(z) = (\alpha + 1)(1 - |z|^2)^\alpha dA(z)$ and let A_α^p be the corresponding analytic L^p space. We use $A^p = A_0^p$. For holomorphic h and $\psi : \mathbb{D} \rightarrow \mathbb{D}$, set

$$W_{h,\psi}f = h(f \circ \psi).$$

The normalized A_α^2 kernel is

$$k_w^\alpha(z) = \frac{(1 - |w|^2)^{(\alpha+2)/2}}{(1 - \bar{w}z)^{\alpha+2}},$$

and the source's normalized A^p tests are

$$\ell_w^{(p)}(z) = \frac{(1 - |w|^2)^{2/p}}{(1 - \bar{w}z)^{4/p}}. \quad (1)$$

2 Results

Theorem 1 (Finite-Blaschke extension). *Let B be a finite Blaschke product.*

1. *Let $1 \leq p < \infty$, let $h \in A^p$, and suppose $W_{h,B} : A^p \rightarrow A^p$ is bounded. With no restriction on the zero set of h , the operator is bounded below if and only if*

$$\inf_{w \in \mathbb{D}} \|W_{h,B} \ell_w^{(p)}\|_{A^p} > 0.$$

2. *Let $\alpha > -1$, let $h \in A_\alpha^2$, and suppose $W_{h,B} : A_\alpha^2 \rightarrow A_\alpha^2$ is bounded. Then it is bounded below if and only if*

$$\inf_{w \in \mathbb{D}} \|W_{h,B} k_w^\alpha\|_{A_\alpha^2} > 0.$$

Thus the two source questions have positive answers for every proper holomorphic self-map of the disk.

Theorem 2 (Compatible infinite zero sets). *Let $1 \leq p < \infty$, let $\psi : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic, and suppose $W_{h,\psi}$ is bounded on A^p . Assume that, with at most finitely many exceptions, every zero a of h satisfies*

$$\frac{p}{2} \text{ord}_a(h) \in \mathbb{N}. \quad (2)$$

Then the conclusion of the source's Theorem 3.4 holds without requiring finitely many zeros: $W_{h,\psi}$ is bounded below if and only if it is uniformly bounded below on the tests (1).

In particular, the conclusion holds for every even integer p and an arbitrary infinite zero set. If $p = 2a/b$ in lowest terms, it holds whenever all but finitely many zero multiplicities are divisible by b .

3 Proof intuition

The pullback measure of a weighted composition operator converts both the operator inequality and the kernel test into measure statements. A general Berezin lower bound is not enough: singular counterexamples are now known [5]. The finite-Blaschke hypothesis supplies the missing regularity. All critical values lie in a compact subdisk, while near the boundary the pullback density is a finite sum of moduli of analytic functions. Its Carleson upper bound and the submean inequality force that density to be bounded. The compact inner part disappears asymptotically from normalized kernels, and a Fredholm argument reduces the problem to Luecking's theorem for bounded densities.

The second theorem uses a different mechanism. Condition (2) is exactly the monodromy condition for $h^{p/2}$ to be analytic. Once $|H|^2 = |h|^p$ with analytic H , the A^p problem is literally the source's established A^2 problem for the same pullback measure.

4 A tail-bounded density lemma

For a positive measure μ that is Carleson for A_α^2 , write

$$\tilde{\mu}_\alpha(w) = \int_{\mathbb{D}} |k_w^\alpha(z)|^2 d\mu(z).$$

Lemma 3 (Tail-bounded Berezin principle). *Suppose $d\mu = g dA_\alpha$, $g \geq 0$, and g is essentially bounded on $\{z : |z| > r_0\}$ for some $r_0 < 1$. If $\inf_{w \in \mathbb{D}} \tilde{\mu}_\alpha(w) > 0$, then μ is a reverse Carleson measure for A_α^2 .*

For $\alpha = 0$, the outer bounded-density part also dominates A^p for every $0 < p < \infty$.

Proof. Choose $r_1 > r_0$ and split

$$\mu = \nu + \sigma, \quad d\nu = \mathbf{1}_{\{|z| > r_1\}} g dA_\alpha.$$

The finite measure σ is supported in a compact subdisk. Since the normalized kernels tend to zero uniformly on compact subsets, $\tilde{\sigma}_\alpha(w) \rightarrow 0$ as $|w| \rightarrow 1$. Hence $\tilde{\nu}_\alpha$ is bounded below for $|w|$ sufficiently close to one.

Add to the density of ν the indicator of a fixed compact disk. The new density q is bounded, and $\widetilde{q, dA_{\alpha\alpha}}$ is bounded below on all of $\mathbb{D}$: near the boundary this comes from ν , and on the remaining compact set it comes from the added disk. The bounded-density Douglas–Luecking theorem says that T_q is invertible. For completeness, its geometric core is short: normalized kernel mass outside a sufficiently large fixed pseudohyperbolic disk is uniformly small; on the disk the kernel density is comparable to reciprocal weighted area. A lower Berezin bound and $q \in L^\infty$ therefore make a level set $\{q > \delta\}$ occupy a fixed positive proportion of every such disk. Luecking’s dominating-set theorem then gives the reverse embedding [3, 4, 5].

The difference $T_q - T_\nu$ has compactly supported bounded symbol and is compact. Thus T_ν is Fredholm. Moreover ν is nonzero, since its Berezin transform is bounded below near the boundary. If $T_\nu f = 0$, then

$$0 = \langle T_\nu f, f \rangle = \int |f|^2 d\nu.$$

A nonzero analytic function cannot vanish on the positive-area set on which the density of ν is positive, so $f = 0$. The self-adjoint Fredholm operator T_ν is therefore invertible. Hence ν , and a fortiori μ , is reverse Carleson.

When $\alpha = 0$, Luecking’s bounded-symbol characterization yields a level set of the density of ν that is dominating. The geometric density condition for that set is independent of p , so the same set dominates every A^p ; this is also the exponent-independence used in the source’s proof of Theorem 3.4. $\square$

5 Proof of Theorem 1

Proof. We give the unweighted A^p argument first. Define

$$\mu(E) = \int_{B^{-1}(E)} |h(z)|^p dA(z). \tag{3}$$

Then

$$\|W_{h,B}f\|_{A^p}^p = \int_{\mathbb{D}} |f|^p d\mu. \tag{4}$$

Boundedness of the operator makes μ a Bergman Carleson measure. Since $|\ell_w^{(p)}|^p = |k_w^0|^2$, the assumed test lower bound is precisely

$$\inf_{w \in \mathbb{D}} \tilde{\mu}_0(w) > 0. \quad (5)$$

The area formula gives, almost everywhere,

$$d\mu(u) = g(u)dA(u), \quad g(u) = \sum_{B(z)=u} \frac{|h(z)|^p}{|B'(z)|^2}. \quad (6)$$

All critical values of B lie in a compact subdisk. Near any sufficiently outer u , choose local inverse branches $\phi_1, \dots, \phi_N$. On a small disk there is a holomorphic branch of $(\phi'_j)^{2/p}$, and hence

$$g(v) = \sum_{j=1}^N \left| (h \circ \phi_j)(v) (\phi'_j(v))^{2/p} \right|^p.$$

Thus g has a locally subharmonic representative in an outer annulus. For a fixed small $c > 0$, the disk $D(u, c(1 - |u|))$ remains in an inverse-branch chart when $|u|$ is sufficiently large. The submean inequality and the Carleson disk estimate give

$$g(u) \lesssim \frac{\mu(D(u, c(1 - |u|)))}{(1 - |u|)^2} \text{lessim} 1.$$

Lemma 3, (5), and its p -independence statement now show that (4) is bounded below. The reverse implication is immediate because the tests have norm one.

For A_α^2 , use

$$\mu_\alpha(E) = \int_{B^{-1}(E)} |h(z)|^2 dA_\alpha(z).$$

Its density relative to dA_α is

$$g_\alpha(u) = \sum_{j=1}^N |h(\phi_j(u))|^2 |\phi'_j(u)|^2 \left(\frac{1 - |\phi_j(u)|^2}{1 - |u|^2} \right)^\alpha \quad (7)$$

in the outer annulus. Because a finite Blaschke product extends across the circle and has nonzero boundary derivative, the derivative and ratio factors in (7) are uniformly bounded above and below on outer Bergman disks. Apply the submean inequality to $|h \circ \phi_j|^2$, then the weighted Carleson disk upper bound, to obtain $g_\alpha(u) \lesssim 1$ near the boundary. Lemma 3 completes the proof. $\square$

6 Proof of Theorem 2

Proof. First remove the finitely many exceptional zeros by the bounded finite contractive divisor used in the source. Multiplication by that divisor is bounded above and below onto its closed finite-codimensional range, so it preserves boundedness below both on the whole space and on the test family. We may therefore suppose (2) holds at every zero.

On the punctured disk, analytically continue $h^{p/2}$. A circuit around a zero of order m multiplies a branch by $e^{2\pi i(pm/2)}$, which equals one by (2). All monodromy is therefore trivial. The resulting single-valued function extends across each zero with order $pm/2$, giving an analytic H such that

$$|H|^2 = |h|^p. \quad (8)$$

In particular $H \in A^2$.

The pullback measure induced by $W_{H,\psi}$ on A^2 is exactly the measure induced by $W_{h,\psi}$ on A^p . The standard forward Bergman Carleson criterion is exponent-independent, so the former operator is bounded. Moreover

$$(\ell_w^{(p)})^{p/2} = k_w^0, \quad \|W_{H,\psi} k_w^0\|_{A^2}^2 = \|W_{h,\psi} \ell_w^{(p)}\|_{A^p}^p.$$

By Theorem 3.1 of the source, this operator is bounded below. If $p = 2m$ is an even integer, the transfer back is completely direct: for every $f \in A^{2m}$, put $F = f^m \in A^2$. Then

$$\|W_{H,\psi} F\|_{A^2}^2 = \|W_{h,\psi} f\|_{A^{2m}}^{2m}, \quad \|F\|_{A^2}^2 = \|f\|_{A^{2m}}^{2m}.$$

For the remaining compatible exponents, the exponent-independent reverse Carleson step used in the source's own proof transfers the common-measure inequality back to A^p . This proves the forward implication; necessity is immediate. $\square$

7 Upgrade attempts and remaining obstruction

Six focused upgrades were audited.

1. A direct arbitrary-measure argument fails: arXiv:2405.05412 constructs a Carleson measure with Berezin transform bounded below but a noninvertible Toeplitz operator [5].
2. Exact scalar or vector Hilbertization cannot cross an incompatible zero multiplicity: the leading vanishing order of a squared holomorphic norm is an even integer, forcing (2).
3. Finite branched covers resolve finitely many rational-power branch points, but infinitely many prescribed branch points require an infinite cover and destroy the finite area-change identity.
4. The tail-density proof extends beyond finite Blaschke products whenever the critical values are compactly confined and outer valence/inverse charts are uniformly controlled.
5. For a general symbol, critical values and valence may accumulate at the boundary. Carleson disk bounds then allow narrow unbounded density spikes, so the decisive outer L^∞ estimate is unavailable.
6. Interpolation between even exponents changes the effective analytic weight; attempting to take fractional powers simply reproduces the monodromy obstruction.

The strongest remaining route is therefore either a new analytic anti-concentration theorem for general weighted pullbacks or a realization of a Douglas-type spiky measure as such a pullback.

8 Verification and novelty

The proof was audited at the area formula, local subharmonicity, uniform outer inverse charts, compact-tail Berezin limit, Fredholm injectivity, and zero-monodromy steps. No computation is used as proof.

The four lightweight run indexes had no entry for the source paper or these extensions. Bounded primary-source searches on 13 August 2026 used its exact title, the two exact open-problem phrases, finite Blaschke symbols, contractive divisors, and lower reproducing-kernel theses. They found the

source and general Berezin/Toeplitz work, but no later paper stating either theorem. Novelty confidence is moderate.

Human-review recommendation. Check first the tail-bounded density lemma’s compact-perturbation step and the uniform weighted comparison in (7). Then check that the source’s exponent-independent reverse Carleson step applies exactly to the common measure in (8).

References

- [1] I. Chalendar and J. R. Partington, *The reproducing kernel thesis for lower bounds of weighted composition operators*, Archiv der Mathematik 113 (2019), 179–187; arXiv:1812.08121.
- [2] P. Duren, D. Khavinson, H. S. Shapiro, and C. Sundberg, *Contractive zero-divisors in Bergman spaces*, Pacific J. Math. 157 (1993), 37–56.
- [3] D. H. Luecking, *Inequalities on Bergman spaces*, Illinois J. Math. 25 (1981), 1–11.
- [4] D. H. Luecking, *Closed-range restriction operators on weighted Bergman spaces*, Pacific J. Math. 110 (1984), 145–160.
- [5] J.-H. Chen, Q. Leng, and X. Zhao, *The Douglas question on the Bergman and Fock spaces*, arXiv:2405.05412 (2024), especially Theorems 3.4 and 3.11.